Recall: phantom category: smooth proper nonzero DG category $\mathcal{P}$, such that $HH_*(\mathcal{P}) = 0$.

$$\mathcal{P} = Perf(\Gamma) / \langle L_1, \cdots, L_{13} \rangle$$

smooth finite dim algebra.

Set $\Delta = k[\varepsilon]/\varepsilon^2$, let $\Gamma_\Delta := \Gamma \otimes_k \Delta$.

$$D_{sg}(\Gamma_\Delta) \simeq K_1(proj(\Gamma)) \simeq D_1(mod(\Gamma)).$$

we can lift this to DG level:

$$sg_{dg}(\Gamma_\Delta) \simeq Perf(Perf(\Gamma)/[1]) \simeq Perf(Perf(\Gamma) \otimes \mathcal{U}).$$

Today: pass to $/\langle L_1, \cdots L_{13} \rangle$.

Def: Let R be a ring, $M \in mod(R)$ is called a generator of $mod(R)$ if $R \in add(M)$.

($\Leftrightarrow$, $\forall N \in mod(R)$ is a quotient of a finite direct sum of M).

Theorem (Xiaowu Chen, A note on the singularity cate --- 2015).

Let R be a finite dim algebra, and $M \in mod(R)$ is a generator. Put $B = End_R(M)$, so $M \in B\text{-mod-}R$, we have a functor

$$- \otimes_B M : mod(B) \longrightarrow mod(R).$$

If $\ker(-\otimes^{\mathbb{L}}_B M) := \{X \in \mathrm{mod}(B) \mid X \otimes^{\mathbb{L}}_B M = 0\} \subseteq \mathrm{Perf}(B)$,

then $-\otimes^{\mathbb{L}}_B M$ induces a triangle equivalence

$$-\otimes^{\mathbb{L}}_B M : D_{sg}(B) \xrightarrow{\sim} D_{sg}(R)/\langle M \rangle.$$

Remarks: Note that $M \in \mathrm{proj}(B)$: since M is a generator of $\mathrm{mod}(R)$, $M^{\oplus n} \simeq R \oplus Q$, then

$M^{\oplus n} \simeq Be$, $e = id_R$

1. $-\otimes_B M = -\otimes^{\mathbb{L}}_B M$.

2. In $D_{sg}(B)$: $M = 0$ (since $M \in \mathrm{proj}(B)$), in order to define functor to $D_{sg}(R)$, we have to quotient $\langle M \rangle$.

3. $D(\mathrm{mod}(B)) \longrightarrow D_{sg}(R)/\langle M \rangle$.

To make sure nothing more in $D(\mathrm{mod}(B))$ is sent to 0, require $\ker(-\otimes^{\mathbb{L}}_B M) \subseteq \mathrm{Perf}(B)$.

Lemma: Chen's partial resolution theorem can be lifted to DG level.

Proof: formal. □

In particular: For $\forall N \in \mathrm{mod}(R)$: set $M = R \oplus N$.

so M is a generator.

$B = \mathrm{End}_R(R \oplus N)$, $e = id_R \in B$, we have

$\ker(_ \otimes N) = \mathrm{mod}(B/e)$. If

Chen's theorem is applicable: (i.e. $\mathrm{mod}(B/e) \subseteq \mathrm{Perf}(B)$)

then $Sg_{dg}(B) \simeq Sg_{dg}(R)/\langle R \oplus N \rangle$

$\simeq Sg_{dg}(R)/\langle N \rangle$.

$Sg_{dg}(\Gamma_\Delta) = \mathrm{Perf}(\mathrm{Perf}(\Gamma)/[1])$

$\mathcal{P} = \mathrm{Perf}(\Gamma)/\langle L_1, \dots, L_{13} \rangle$.

Want algebra A: $Sg_{dg}(A) \simeq Sg(\Gamma_\Delta)/\langle F\text{-}id(L_1), \dots, F\text{-}id(L_{13}) \rangle$

$\simeq \mathrm{Perf}(\mathrm{Perf}(\Gamma)/[1])/\langle L_1, \dots, L_{13} \rangle$.

Choose $L_i \in C(\mathrm{mod}(\Gamma))$ to be projective representatives. Take $N_i = (L_i \in C_1(\mathrm{mod}\,\Gamma))$

N_i

$(F\text{-}id(L_i) \in \mathrm{mod}(\Gamma_\Delta))$

Note that $N_i \in proj(\Gamma)$.

Construction of A:

Def: Set $A = \mathrm{End}_{\Gamma_\Delta}(\Gamma_\Delta \oplus N_1 \oplus N_2 \oplus \dots \oplus N_{13})$

$e = id_{\Gamma_\Delta} \in A$.

Proposition: Chen's theorem applies for A.

$$Sg_{dg}(A) \simeq Sg_{dg}(\Gamma_\Delta)/\langle Fold(L_1), \cdots, Fold(L_{13})\rangle$$

Proof: Set $M = \Gamma_\Delta \oplus (N_1 \oplus \cdots \oplus N_{13})$

$$N = N_1 \oplus \cdots \oplus N_{13}.$$

To apply Chen's theorem: need to show

$$\ker(- \otimes_A M) = mod(A/e) \subseteq Perf(A).$$

Note that $A/e \cong \underline{Hom}_{\Gamma_\Delta}(N, N)$:

any element in (e) is a linear combination of maps that factor through Γ_Δ. Conversely, $\forall f \in \underline{Hom}_{\Gamma_\Delta}(N,N)$ can be lifted to a map that factors through $\Gamma_1^{\oplus n}$, so it lies in (e).

We have:

$$A/e \cong \underline{End}_{\Gamma_\Delta}(N) \cong End_{H^0(Perf(\Gamma)/[1])}\left(\bigoplus_{i=1}^{13} L_i\right)$$

$$= \bigoplus_{j \in \mathbb{Z}} Hom_{Perf(\Gamma)}\left(\bigoplus_{i=1}^{13} L_i, \bigoplus_{i=1}^{13} L_i[j]\right).$$

Since $\langle L_1, \cdots, L_{13}\rangle$ is an exceptional collection

So we can see A/e is smooth so

$$mod(A/e) \subseteq Perf(A/e).$$

Claim: we have exact complex:

summand of $\mathrm{Hom}(M, M^{\oplus m})$ $\quad N \in \mathrm{add}(M)$, $N \otimes \Gamma_\Delta \in \mathrm{add}(M)$

$$0 \to \mathrm{Hom}'_{\Gamma_\Delta}(M,N) \to \mathrm{Hom}_{\Gamma_\Delta}(M, N \otimes \Gamma_\Delta) \qquad A = \mathrm{End}_{\Gamma_\Delta}(M)$$

$$\to \mathrm{Hom}_{\Gamma_\Delta}(M,N) \to \underbrace{\mathrm{Hom}_{\Gamma_\Delta}(M,N)/(e)}_{\cong} \qquad A/e.$$

$$\mathrm{Hom}_{\Gamma_\Delta}(N,N) \to 0$$

Exactness of first 2 nonzero terms:

$0 \to N \to N \otimes \Gamma_\Delta \to N \to 0$, applying $\mathrm{Hom}_{\Gamma_\Delta}(M, -)$ gives left-exactness.

For exactness at $\mathrm{Hom}_{\Gamma_\Delta}(N,N)$ (3rd term):

if $f \in \mathrm{Hom}(M,N)$ is mapped to $0 \in \mathrm{Hom}(N,N)$,

then it factors through a projective $P \in \mathrm{proj}(\Gamma_\Delta)$

$f: M \to P \to N$. since $N \in \mathrm{proj}(\Gamma)$,

$N \otimes \Gamma_\Delta \in \mathrm{proj}(\Gamma_\Delta)$, we have $(N \otimes \Gamma_\Delta)^{\oplus n} \to P$.

so $M \to P$ lifts to $(N \otimes \Gamma_\Delta)^{\oplus n}$, hence

$\ker\left(\mathrm{Hom}(M,N) \to \mathrm{Hom}(N,N)\right) =$ image of

$\mathrm{Hom}(M, N \otimes \Gamma_\Delta)$.

Surjective at $\underline{\mathrm{Hom}}(N, N)$; check. ✓

The first 3 terms are $\in \mathrm{proj}(A)$, so this gives a finite projective A-resolution of A/e.

Hence $A/e \in \mathrm{Perf}(A)$,

$$\mathrm{mod}(A/e) \subseteq \mathrm{Perf}(A/e) \subseteq \mathrm{Perf}(A)$$

So condition in Chen's theorem is satisfied. □

Lemma:

$$\mathrm{Perf}(\mathrm{Perf}(\Gamma)/[1])/\langle L_1, \dots, L_{13}\rangle$$

$$\simeq \mathrm{Perf}\left(\left(\mathrm{Perf}(\Gamma)/\langle L_1, \dots, L_{13}\rangle\right)/[1]\right)$$

Proof: formal. □ P.

So: $\mathrm{Sg}_{dg}(A) \simeq \mathrm{Perf}(\mathcal{P}/[1])$

$$\simeq \mathrm{Perf}(\mathcal{P} \otimes \mathcal{Q})$$

Lemma: Let $\mathcal{C}, \mathcal{D}$ be DG categories over a field k. Then

$$HH_*(\mathcal{C} \otimes_k \mathcal{D}) \cong HH_*(\mathcal{C}) \otimes_k HH_*(\mathcal{D}).$$

Proof: [Dmitry Shklyarov, HRR for DG algebras 2013]. □

This lemma is called Hochschild Künneth theorem.

Hence: $HH_*(sg(A)) \simeq HH_*(\mathcal{P} \otimes \mathcal{Q})$

$= (HH_*(\mathcal{P}) = 0) \otimes HH_*(\mathcal{Q}) = 0$.

Recall: $HH_n(sg_{dg}(A)) \cong HH_{n-1}(A)$, $n \geq 2$.

If we want A, $HH_{\geq 1}(A) = 0$, it suffices to produce $sg(A)$ s.t. that $HH_*(sg_{dg}(A)) = 0$

By above argument: we have such A, $D_{sg}(A) \neq 0$, $HH_*(sg(A)) = 0$, and A is finite dim.

So Han's Conj is false. □

folding in lecture 3.

$$\begin{array}{ccc} Perf(Perf(\Gamma)/\langle\square\rangle) & \xrightarrow{\sim} & sg(\Gamma_\Delta) \\ \downarrow & & \downarrow \\ Perf(Perf(\Gamma)/\langle\square\rangle/\langle L_1,\dots,L_{43}\rangle) & \xrightarrow{\sim} & sg(\Gamma_\Delta)/\langle L_1,\dots,L_{43}\rangle \end{array}$$

Chen's theorem

$\Rightarrow sg(A)$

S1

$Perf((Perf(\Gamma)/\langle L_1,\dots L_{43}\rangle)/(1))$

$\mathcal{P}^{11}$, phantom category.